Gatlin A

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PROFILE

Computer Science undergraduate specializing in Machine Learning, Deep Learning, and AI Engineering, with hands-on experience building neural networks and LLMs from scratch, LLM inference optimization, fine-tuning, LLM-based agents, and anomaly detection. Strong in Python and PyTorch, with systems work in Rust and Go.

EDUCATION

Karunya University, Tamil Nadu — B.Tech, Computer Science and Engineering 2024 – Present

Good Shepherd Model School, Tuticorin — Higher Secondary (Computer Science & Mathematics) 2022 – 2024

TECHNICAL SKILLS

Languages	Python, Rust, Go, Java, C, TypeScript
Machine Learning	PyTorch, Deep Learning, NLP, Neural Networks, Computer Vision, Anomaly Detection, Drift Detection
LLM / AI	LLMs, LLM Inference (KV cache, batching, quantization), LoRA Fine-tuning, Distillation, AI Agents, Tokenization, Embeddings, Attention
Backend	FastAPI, Flask, Node.js, REST APIs, SSE Streaming
Databases	PostgreSQL, MySQL, Redis, SQLite, Firebase
Tools	Git, GitHub, Docker, Linux, VS Code
Core CS	Data Structures & Algorithms, OOP, DBMS, Storage Engines

EXPERIENCE

Web Development Intern — D Data Technologies June 2026

- Completed a 4-week internship contributing to web application development tasks, applying programming, debugging, and API integration practices in a professional environment.

PROJECTS

Pagewise — Mini LLM Inference Engine (vLLM-style) [GitHub] *Python, PyTorch, FastAPI*

- Built an LLM inference engine from scratch in PyTorch (own Qwen2 model code, generation loop and scheduler, verified against Hugging Face): KV cache (4.0x decode at 2K context), continuous batching (10.7x throughput) and a paged KV cache fitting 71 vs 21 sequences in the same memory.
- Added int8/int4 weight quantization (942 to 472 MiB, equal perplexity) and an OpenAI-compatible SSE server serving 578 tok/s to 32 streaming clients on an 8GB laptop GPU.

Azii — Custom AI Language Model [GitHub] *Python, PyTorch, NLP, FastAPI*

- Built a ~27M-parameter GPT-style model from scratch with a custom byte-level BPE tokenizer; pretrained on ~485M TinyStories tokens and chat-finetuned on SODA with loss masking, all on an 8GB GPU, with a streaming FastAPI + React chat UI.

Axii — Offline Emotional-Companion Chatbot [GitHub] *Python, PyTorch, LoRA, SQLite*

- Trained a 4.2M-parameter emotion Transformer from scratch (GoEmotions, macro-F1 0.445) to steer a LoRA fine-tuned Qwen2.5-1.5B; distilled 3.1k replies from a teacher model, lifting a 39-check behavioural eval from 8/39 to 30/39. Long-term SQLite memory and a crisis-safety layer.

Harbinger — Anticipatory ML Drift Response [GitHub] *Python, scikit-learn, river*

- Forecasts harmful distribution shift and chooses retrain / replay / rollback; evaluated threshold, expected-cost and LinUCB policies against reactive ADWIN over 50 seeds with paired bootstrap CIs and Elec2 / INSECTS real data.

Autonomous LLM Agent & Tool-Use Framework [GitHub] *Python, Qwen, Node.js, REST APIs*

- LLM agent turning high-level goals into validated JSON actions, with planning, short/long-term memory and failure recovery; local Qwen model via an OpenAI-compatible API and an execution-feedback loop that reduced invalid and repeated actions.

Cynova — Adaptive Forward Validation for Anomaly Detection *Python, PyTorch, Autoencoder*

- Autoencoder on ~1 year of multivariate NOAA ASOS weather data with an Adaptive Forward Validation framework (future-evidence windows, persistence thresholds) separating transient anomalies from persistent events.

Real-Time Footfall Detection & Tracking *Python, PyTorch, YOLO11, BoT-SORT*

- Trained YOLO11s on MOT17 (13,098 images): 0.917 precision, 0.836 mAP@50; added tracking, appearance Re-ID and line-crossing logic for directional counting.

Ismkv — LSM-Tree Key-Value Store [GitHub] *Go (stdlib only)*

- LevelDB-style engine with WAL crash recovery, SSTables, bloom filters, leveled compaction, MVCC snapshots and transactions; random reads 191k to 460k ops/s (+150%) after adding a block cache.

Kaalam — Game Boy Emulator with Rewind [GitHub] *Rust*

- Emulator with rewind, deterministic record/replay and a hardware inspector; passes all 500k SingleStepTests CPU cases, Blargg cpu_instrs and 62/66 Mooneye tests.

School ERP Management System *FastAPI, MySQL, Redis, Docker*

- Layered ERP backend with JWT auth, role-based access across 11 roles, MFA/OTP, refresh tokens, rate limiting and audit logging.

CERTIFICATIONS

- Intermediate Machine Learning — Kaggle (Aug 2026)
- Introduction to Modern AI — Cisco Networking Academy (Aug 2026)
- AI Fundamentals: Foundations for Understanding AI — Cisco Networking Academy (Sep 2026)
- CCNA: Enterprise Networking, Security, and Automation — Cisco Networking Academy